CUSTOMER SHOPPING BEHAVIOR PROJECT

USING PYTHON, SQL AND POWER BI

Python
Data Cleaning & Preparation

SQL
Data Analysis & Insights

Power BI
Interactive Dashboard & Visualization





- **Customer Insights**
- **Product Performance**
- **Revenue Trends**
- **Business Recommendations**

PROJECT OVERVIEW

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01



Business Problem

02



Python —
Data Cleaning &
Preparation

03



SQL —
Business Analysis
& Insights

04



Power BI —
Dashboard &
Visualization

05



Insights &
Recommendations

06



Documentation —
Project Report &
Presentation

07



GitHub —
Code & Project Files

BUSINESS PROBLEM STATEMENT

A retail company wants to understand customer shopping behavior to improve sales, customer engagement, customer loyalty, and marketing effectiveness.

The company wants to identify how customer demographics, product preferences, discounts, shipping methods, reviews, and subscription status relate to purchasing behavior and revenue.

Main business question

How can the company use customer shopping data to identify purchasing trends, improve customer engagement, and optimize marketing and product strategies?

PROJECT DELIVERABLES

1. Data Cleaning & Preparation (Python)
Clean and transform the raw dataset, handle missing values, standardize column names, and prepare data for analysis.

2. Business Data Analysis (SQL)
Analyze customer shopping behavior using SQL queries to uncover insights into revenue, product performance, discounts, and customer loyalty.

3. Dashboard & Visualization (Power BI)
Build an interactive dashboard to visualize key metrics, customer trends, and business performance.

4. Business Insights & Recommendations
Summarize key findings and recommend actionable strategies to improve customer engagement, marketing, and sales.

5. Project Documentation & GitHub
Organize the Python notebook, SQL queries, Power BI dashboard, and project documentation for portfolio presentation.

Understand Data



customer_shopping_behavior

DATA CLEANING & PREPARATION (PYTHON)

1. **Missing Value Treatment** — Imputed 37 missing review ratings using the median rating of each product category.
2. **Column Name Standardization** — Converted column names into lowercase snake_case format for consistency.
3. **Feature Engineering** — Created an age_group column to categorize customers into four age groups.
4. **Data Transformation** — Converted purchase frequency categories into numeric day values.
5. **Redundant Column Removal** — Removed promo_code_used after confirming it matched discount_applied across all records.
6. **Data Validation** — Verified that no missing values remained after the cleaning process.

DATA ANALYSIS (SQL)

1. **Revenue Analysis** — Compared total revenue generated by male and female customers.
2. **Discount & Spending Analysis** — Identified discount users whose purchase amounts were at or above the overall average.
3. **Product Rating Analysis** — Ranked the top 5 products by average review rating.
4. **Shipping Analysis** — Compared average purchase amounts for Standard and Express shipping.
5. **Subscription Analysis** — Compared customer counts, average spending, and total revenue between subscribers and non-subscribers.
6. **Discount Dependency Analysis** — Identified the top 5 products by percentage of purchases with discounts applied.
7. **Customer Segmentation** — Classified customers into New, Returning, and Loyal segments based on previous purchases.
8. **Product Popularity Analysis** — Identified the top 3 most purchased products within each category.
9. **Repeat Buyer Analysis** — Compared subscription status among customers with more than 5 previous purchases.
10. **Age Group Revenue Analysis** — Compared total revenue across customer age groups.

POWER BI DASHBOARD

1. Key Performance Indicators (KPIs)

- Total Number of Customers
- Total Purchase Revenue
- Average Purchase Amount
- Average Review Rating

2. Data Visualizations

- Customer Distribution by Subscription Status
- Revenue by Product Category
- Sales by Age Group
- Purchase Amount by Gender
- Top 10 Products by Revenue

3. Interactive Slicers

- Subscription Status
- Gender
- Product Category
- Shipping Type
- Payment Method

Customer Shopping Behavior Dashboard

No | Yes

Female | Male

Category

Accessories

Clothing

Shipping Type

2-Day Shipping

Express

Free Shipping

payment_method

Bank Transfer

Cash

Credit Card

Clear all slicers

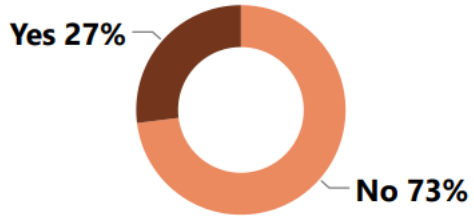
3.9K
Total Customers

99K
Total_Revenue

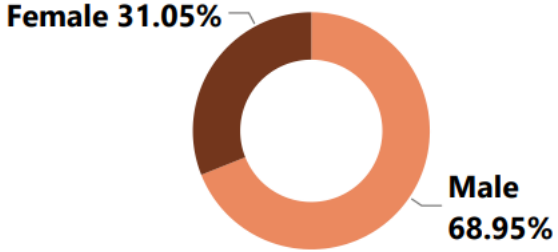
$59.76
Avg Purchase Amt

3.75
Avg Review Rating

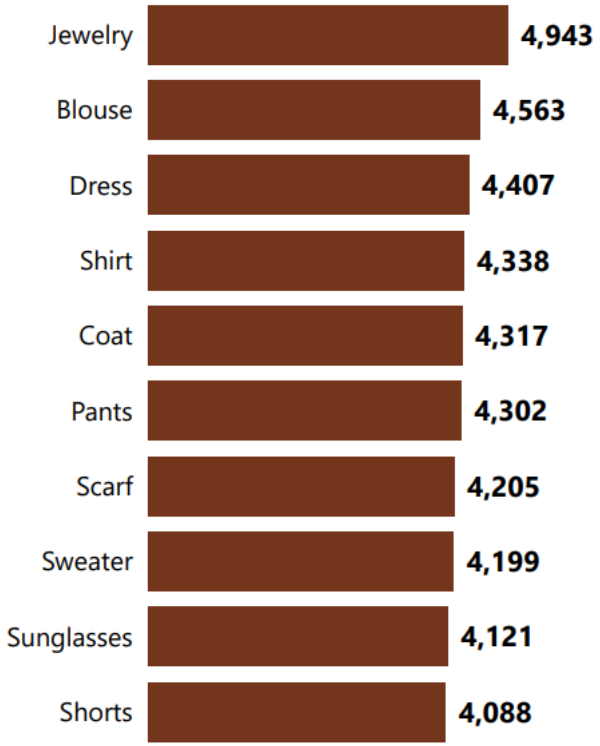
Customers Distribution by Subscription Status



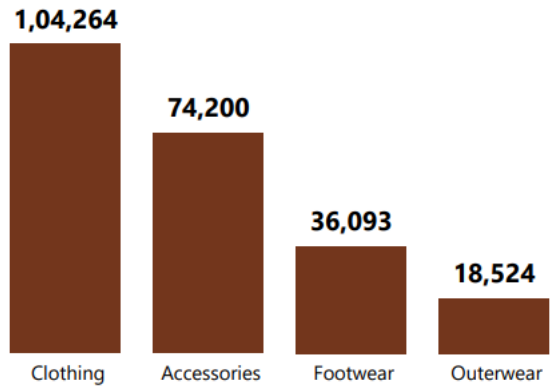
Revenue by Gender



Top 10 Products by Revenue

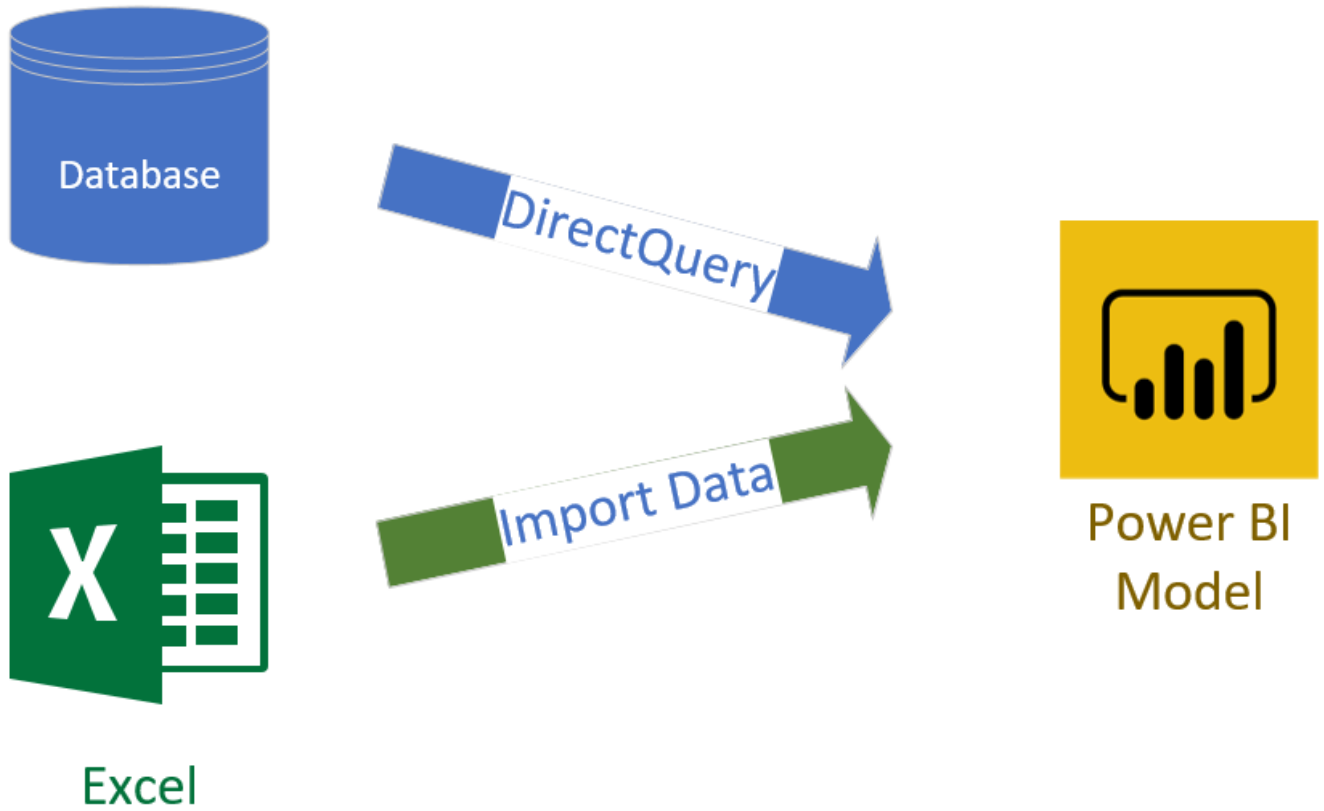


Revenue by Category



Sales by Age Group




Database
DirectQuery
Import Data
Power BI
Model
Excel

Customer Shopping Behavior Dashboard

No | Yes

Female | Male

Category

- Accessories
- Clothing
- Footwear

Shipping_Type

- 2-Day Shipping
- Express
- Free Shipping

Payment_Method

- Bank Transfer
- Cash
- Credit Card

Reset

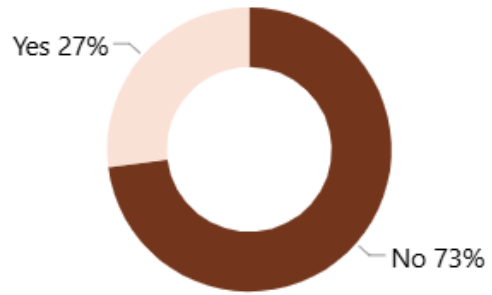
3,900
Total Customer

$2,33,081
Total Revenue

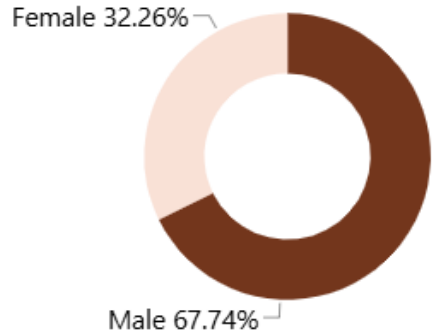
$59.76
Avg Purchase Amount

3.75
Avg Review Rating

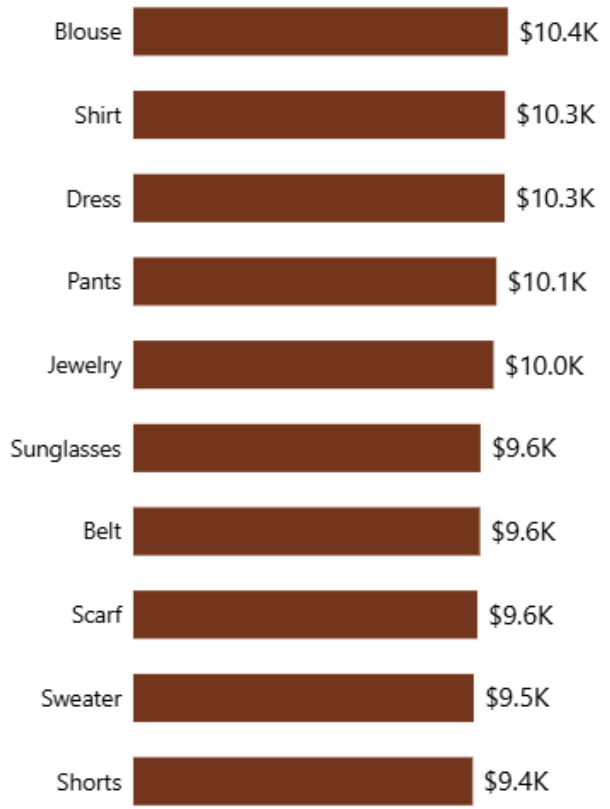
Customers Distribution by Subscribers Status



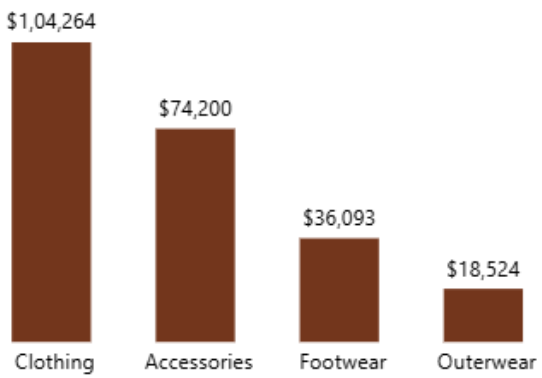
Revenue by Gender



Top 10 Products by Revenue



Revenue by Category



Revenue by Age Group





Documentation — Project Report & Presentation



Insights & Recommendations



GitHub —
Code & Project Files